Introduction to Probability and Statistics

(EE 354 / CE 361)

Practice Problems
Unit 04



Exercise No 1

A Random Variable Z_n is defined as $Z_n = min(X_1; X_2; \ldots; X_n)$, where $X_i's$ are outcomes of identical and independently distributed fair six-sided die. As an example $Z_2 = min(X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of six-sided die X_1 and X_2.

(a) (**5 marks**) Calculate probability $P(1 < Z_{10} < 6)$.

(b) (**5 marks**) Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$



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(a) (**5 marks**) Calculate probability $P(1 < Z_{10} < 6)$.

(a) $P(1 < Z_{10} < 6) = 1 - P(Z_{10}=1) - P(Z_{10}=6)$

$P(Z_{10}=1) = 1 - P(Z_{10} \neq 1)$

$= 1 - P(Z_{10} \geq 2)$

$= 1 - P(\text{All 10 die are between 2-6})$

$P(Z_{10}=1) = 1 - \left(\frac{5}{6}\right)^{10}$

$P(Z_{10}=6) = P(\text{All 10 die are 6})$

$= \left(\frac{1}{6}\right)^{10}$

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$= \left(\frac{1}{6}\right)^{10}$

$P(1 \leq Z_{10} < 6) = 1 - \left(1 - \left(\frac{5}{6}\right)^{10}\right) - \left(\frac{1}{6}\right)^{10}$

$= 1 - 1 + \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10}$

$= \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10}$ — 0.1615



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(a) (**5 marks**) Calculate probability $P(1 < Z_{10} < 6)$.

(b) (**5 marks**) Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$

$P(Z_n = 1) = 1 - P(Z_n \neq 1)$

$= 1 - P(\text{All } n \text{ throws are greater than } 1)$

$= 1 - \left(\frac{5}{6}\right)^n$

$\Rightarrow 1 - \left(\frac{5}{6}\right)^n \geq 0.6$

$\left(\frac{5}{6}\right)^n \leq 0.4 \Rightarrow n \log \frac{5}{6} \leq \log 0.4$

$n \geq \frac{-0.397}{-0.079}$

$n \geq 5.02$

$n = 6$

Question 1 - (15 points)

Let's be optimistic and assume that Pakistan cricket team has qualified for the T20 World Cup Semi Final. Pakistan's opponent in Semi Final will be West Indies, England, or Australia with a probability of 0.4, 0.4, and 0.2 respectively. The probability of Pakistan winning against West Indies is 0.6. The probability of Pakistan winning against England is 0.4. The probability of Pakistan winning against Australia is 0.6.

a) (7.5 points) What is the probability that Pakistan will win the Semi-final.

b) (7.5 points) Assume that you decided to completely switch off from cricket and did not follow the game. Later, you were told that Pakistan lost the Semi Final. What is the probability that Pakistan played the Semi Final game against England?

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Solution:

(a)

Let's assume the following naming convantions for various events :

A_1 = {Pakistan play West Indies in Semi Final}

A_2 = {Pakistan play England in Semi Final}

A_3 = {Pakistan play Australia in Semi Final}

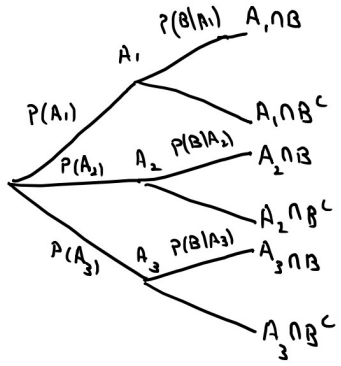
B = {Pakistan win the Semi Final}

B^c = {Pakistan lose the Semi Final}

$\Rightarrow \quad P(A_1) = 0.4 \qquad P(A_2) = 0.4 \qquad P(A_3) = 0.2$

$P(B|A_1) = 0.6 \qquad P(B|A_2) = 0.4 \qquad P(B|A_3) = 0.6$

$P(B^c|A_1) = 0.4 \qquad P(B^c|A_2) = 0.6 \qquad P(B^c|A_3) = 0.4$



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By Total Probability Theorem:-

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= (0.4)(0.6) + (0.4)(0.4) + (0.2)(0.6)$$

$$= 0.24 + 0.16 + 0.12$$

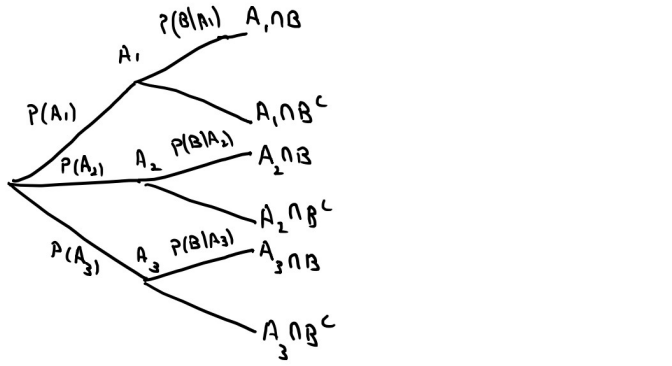
$$P(B) = 0.52 \text{ And}$$

$$P(B^c) = 1 - P(B) = 0.48$$

$$\Rightarrow \quad P(A_1) = 0.4 \qquad P(A_2) = 0.4 \qquad P(A_3) = 0.2$$

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$P(B^c) = 1 - P(B) = 0.48$

According to Bayes' Rule:

$$P(A_2|B^c) = \frac{P(A_2)P(B^c|A_2)}{P(B^c)}$$

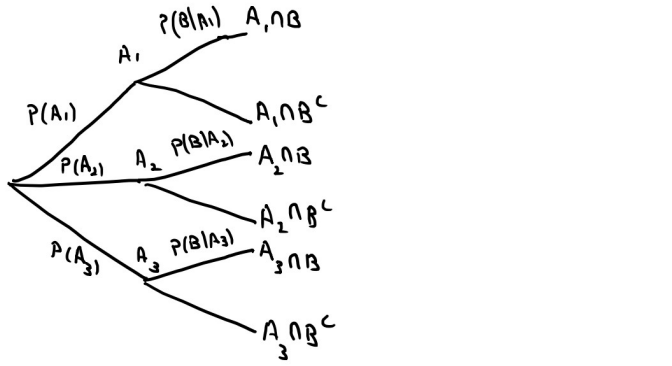
$$= \frac{(0.4)(0.6)}{0.48}$$

$P(A_2|B^c) = 0.5$ — Ans (b)

$\Rightarrow$ $P(A_1) = 0.4$ $P(A_2) = 0.4$ $P(A_3) = 0.2$

$P(B|A_1) = 0.6$ $P(B|A_2) = 0.4$ $P(B|A_3) = 0.6$

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Question 3 - (14 points)

Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

(If the first roll results in 1 and second roll results in 3, the value taken by X is 2)

(If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

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d) (4 points) Calculate Variance of Y

Possible magnitudes of the difference between two rolls:

Roll 2 = 3	2	1	0
Roll 2 = 2	1	0	1
Roll 2 = 1	0	1	2
	Roll 1 = 1	Roll 1 = 2	Roll 1 = 3

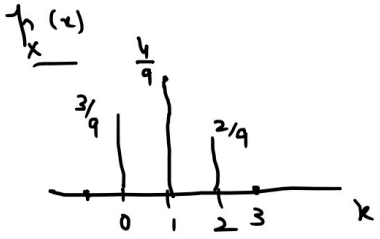
(a)

Possible values that X can take = $\{0, 1, 2\}$

$P(X=0) = \frac{3}{9}$

$P(X=1) = \frac{4}{9}$

$P(X=2) = \frac{2}{9}$



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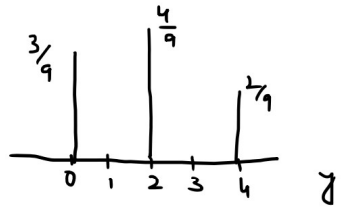
d) (4 points) Calculate Variance of Y

(b)

$Y = 2X$

Possible values of $Y = \{0, 2, 4\}$

$p_Y(y)$:



(c)

$$\mu_Y = E[Y] = \sum_y y\, p_Y(y)$$

$$= 0\left(\frac{3}{9}\right) + 2\left(\frac{4}{9}\right) + 4\left(\frac{2}{9}\right)$$

$$\mu_Y = E[Y] = \frac{16}{9} = 0.17$$

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c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

(d)

$$\text{Var}(Y) = E[Y^2] - (\mu_Y)^2$$

$$E[Y^2] = \sum_y y^2 f_Y(y) \quad \text{(Expected Value Rule)}$$

$$= 0^2\left(\frac{3}{9}\right) + 2^2\left(\frac{4}{9}\right) + 4^2\left(\frac{2}{9}\right)$$

$$= \frac{16}{9} + \frac{32}{9}$$

$$E[Y^2] = \frac{48}{9}$$

$$\text{Var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2$$

$$\text{Var}(Y) = 2.17$$

Question 4 - (4 points)

Consider two random variables X & Y:

X = Temperature in Karachi on Oct 29, 2021 in Celsius (℃)

Y = Temperature in Karachi on Oct 29, 2021in Fahrenheit (℉)

Assume that you have been provided the PMF of X and the corresponding PMF of Y.

Case 1: You take the mean of X and then convert this mean value to ℉

Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $℉ = \mathbf{1.8}℃ + \mathbf{32}$ ***)***

Question 4 - (4 points)

Consider two random variables X & Y:

X = Temperature in Karachi on Oct 29, 2021 in Celsius (°C)

Y = Temperature in Karachi on Oct 29, 2021in Fahrenheit (°F)

Assume that you have been provided the PMF of X and the corresponding PMF of Y.

Case 1: You take the mean of X and then convert this mean value to °F

Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $°F = 1.8°C + 32$)

If Y is a linear function of X, i.e.

$$Y = aX + b$$

$$E[Y] = aE[X] + b$$

Here,

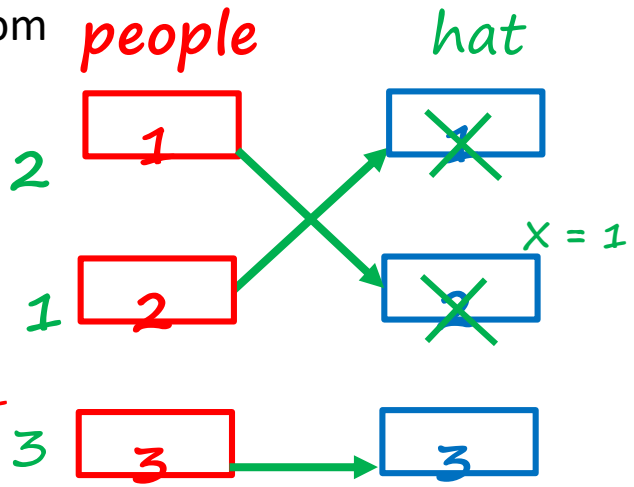
$$Y = 1.8X + 32$$

$$\Rightarrow \underbrace{E[Y]}_{\text{Case 2}} = \underbrace{1.8\,E[X] + 32}_{\text{Computation in Case 1}}$$

Therefore, you will get the same result in both cases.

End of Chapter Exercise – The hat problem

- *n* people throw their hats in a box and then pick one at random
  - ❖ All permutations equally likely $1/n!$
  - ❖ Equivalent to picking one hat at a time



- *X*: number of people who get their own hat
  - ❖ Find **E**[X] $= E[X_1] + \ldots + E[X_n] = n * 1/n = 1$

$$X_i = \begin{cases} 1, & \text{if i selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$$\boldsymbol{X = X_1 + \ldots + X_n}$$

❖ **E**[X_i] = $E[X_1] = P(X_1 = 1) = 1/n$

$\frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{3!}$

$P_X(k)$: hard!

$\sum_k k\, p_X(x)$

The variance in the hat problem

- X: number of people who get their own hat
  - ❖ Find var(X)

$$X_i = \begin{cases} 1, & \text{if i selects own hat} \\ 0, & \text{otherwise} \end{cases}$$

$n = 2$

$X_1 = 1 \Rightarrow X_2 = 1$

$X_1 = 0 \Rightarrow X_2 = 0$

$$X = X_1 + \ldots + X_n$$

$$X^2 = \overbrace{\sum_i X_i^2}^{n} + \overbrace{\sum_{i,j:\, i \neq j} X_i X_j}^{n(n-1)}$$

- $var(X) = E[X^2] - (E[X])^2 = 2 - 1 = 1$

- $E[X_i^2] = E[X_1^2] = E[X_1] = 1/n$

$$\Rightarrow E[X^2] = n\frac{1}{n} + n(n-1)\frac{1}{n} * \frac{1}{n-1}$$

- for $i \neq j$: $E[X_i X_j] = E[X_1 X_2] = P(X_1 X_2 = 1) = P(X_1 = 1, X_2 = 1)$

$$\Rightarrow = P(X_1 = 1) \cdot P(X_2 = 1 \mid X_1 = 1) = \frac{1}{n} * \frac{1}{n-1}$$